Group 1 Assignment

Modems & RS-232 Interfacing Sequences and Line Coding & Its Techniques

CIT 3212 — Computer Networks

Topic area: The Physical Layer (OSI Layer 1)

Prepared in answer to "Assignment on Physical Layer (Group 1)"

Outline

1. **Introduction** — Physical Layer context
2. **Modems** — purpose & operation
3. Modem types & modulation techniques
4. **RS-232** — interface & voltage levels
5. RS-232 pin-out & handshaking sequence
6. **Line coding** — definition & design goals
7. Line coding schemes & waveforms
8. Worked example & comparison
9. Summary

Where this fits

Both topics live in the Physical Layer (Layer 1).

Modems/RS-232 = how two devices exchange data.

Line coding = how bits become a signal.

1. Introduction

Both parts of this assignment belong to the Physical Layer (Layer 1) of the OSI model — the layer that moves raw bits across a transmission medium.

Modems & RS-232

Modem converts digital data into signals suitable for a line.

RS-232 defines the electrical + procedural interface between a computer and a modem/serial device.

Together: how data is carried between two devices.

Line Coding

Converts a sequence of bits into a digital signal.

Placed directly on the baseband medium.

Concerned with signal shape, synchronisation and DC.

Happens before carrier modulation.

2. Modems — Definition & Purpose

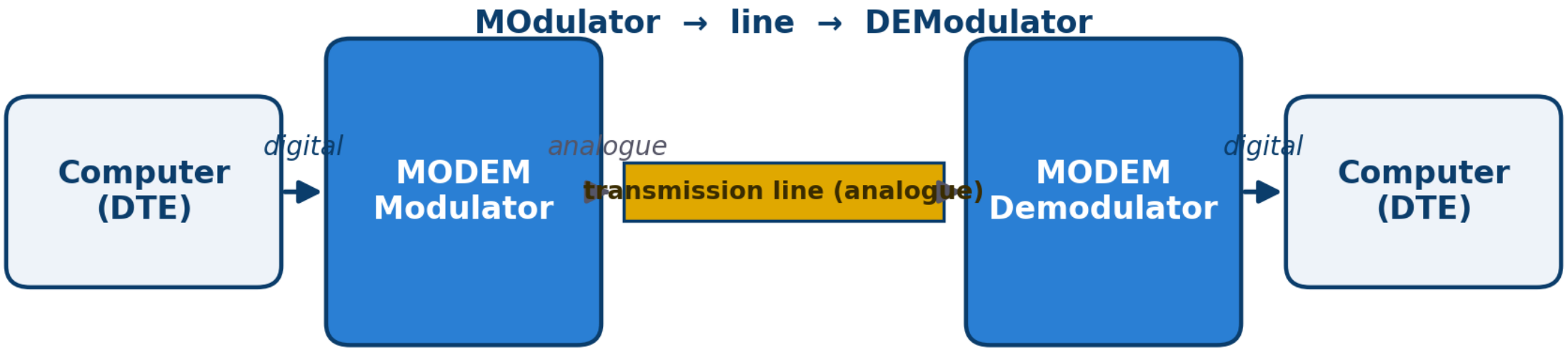
A modem (MOdulator–DEModulator) converts digital data from a computer into an analogue signal for transmission over a line, and converts received analogue signals back into digital data.

- **Modulation** — sender changes a carrier wave to represent 0s and 1s.
- **Demodulation** — receiver extracts the original bits from the carrier.
- Most modems are full-duplex: modulate and demodulate simultaneously.
- Classic Layer-1 device; bridges digital computers and analogue media.
- Carrier is a sine wave whose amplitude, frequency or phase can be varied.

2. How a Modem Works

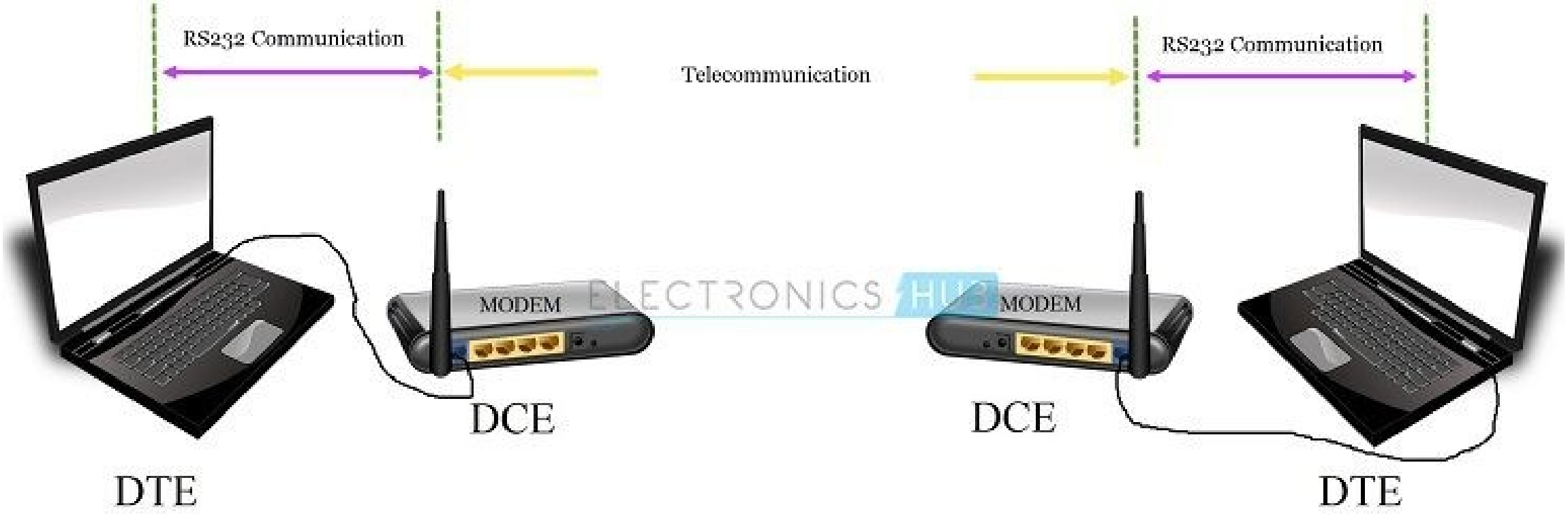
1. Bits from the computer arrive over the serial interface (RS-232).
2. The modulator impresses the bits on a carrier (ASK/FSK/PSK/QAM).
3. The analogue signal travels over the transmission line.
4. The far-end demodulator recovers the bits from the carrier.
5. Bits are passed to the receiving computer via its serial interface.

2. Modem — Block Diagram



Digital data is modulated onto the line and demodulated at the far end.

2. Modem — RS-232 to Laptop (Example)



A laptop connected to a modem through an RS-232 serial cable.

3. Types of Modems

By	Type	Description
Direction	Half-duplex	Both directions, but not at the same time
Direction	Full-duplex	Both directions simultaneously
Medium	Dial-up (PSTN)	Ordinary phone lines; up to 56 kbps
Medium	DSL / ADSL	High-speed data over phone lines, voice unaffected
Medium	Cable modem	Data over cable-TV (coaxial) networks
Connection	Internal / External	Expansion card / separate box via serial cable
Carrier	Wireless / cellular	Uses radio (GSM / 4G / 5G)

3. Modulation Techniques

ASK — Amplitude Shift Keying

Carrier amplitude switched between levels.

FSK — Frequency Shift Keying

Different frequencies represent the bits.

PSK — Phase Shift Keying

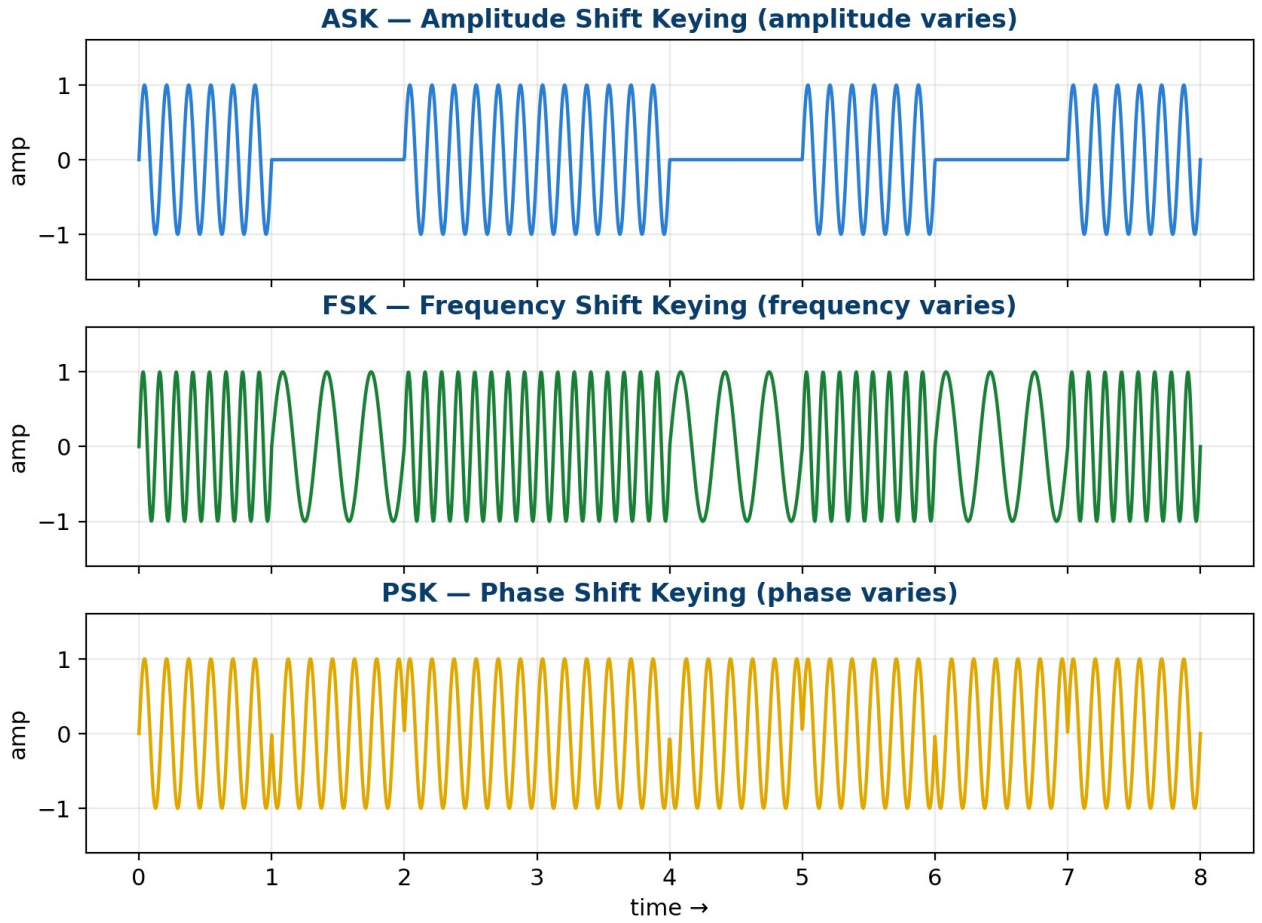
Carrier phase is shifted to encode bits.

QAM — Quadrature Amplitude Modulation

Amplitude + phase combined; many bits per symbol.

bit rate = baud rate × bits per symbol

3. Modulation Waveforms (ASK / FSK / PSK)



Data bits 1 0 1 1 0 1 0 1 encoded by changing amplitude, frequency or phase.

4. RS-232 — What It Is

RS-232 (Recommended Standard 232 / EIA-232) is a standard for serial binary data interchange between Data Terminal Equipment (DTE) and Data Communication Equipment (DCE) over a single-ended, point-to-point link.

It defines three things

Electrical — voltage levels and signalling.

Mechanical — connectors (DB-9 / DB-25) and pin-outs.

Procedural — the sequence of control signals.

DTE vs DCE

DTE = Data Terminal Equipment (computer / terminal).

DCE = Data Communication Equipment (modem).

The interface lets the DTE talk to the DCE.

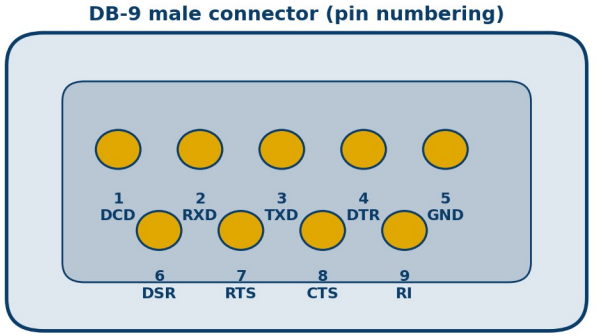
4. RS-232 — Electrical Characteristics

Logic	Signal	Voltage at transmitter
0 (Space)	ON / active	+3 V to +15 V (typ. +12 V)
1 (Mark)	OFF / idle	−3 V to −15 V (typ. −12 V)

- **Negative logic** — large voltage swings resist noise.
- Voltages between −3 V and +3 V are undefined (rejected as noise).
- Maximum cable length ≈ 15 m (50 ft) at 19.2 kbps.
- Common data rates up to 20 kbps (115.2 kbps over short cables).
- Single-ended (unbalanced) signalling → limited noise immunity vs RS-422/RS-485.

4. RS-232 — Pin-Out (DB-9)

Pin	Name	Dir.	Function
1	DCD — Carrier Detect	DCE→DTE	Carrier detected
2	RXD — Receive Data	DCE→DTE	Data received
3	TXD — Transmit Data	DTE→DCE	Data sent
4	DTR — Terminal Ready	DTE→DCE	Computer ready
5	GND — Signal Ground	—	Common reference
6	DSR — Data Set Ready	DCE→DTE	Modem ready
7	RTS — Request To Send	DTE→DCE	Ask to send
8	CTS — Clear To Send	DCE→DTE	Permission to send
9	RI — Ring Indicator	DCE→DTE	Incoming call

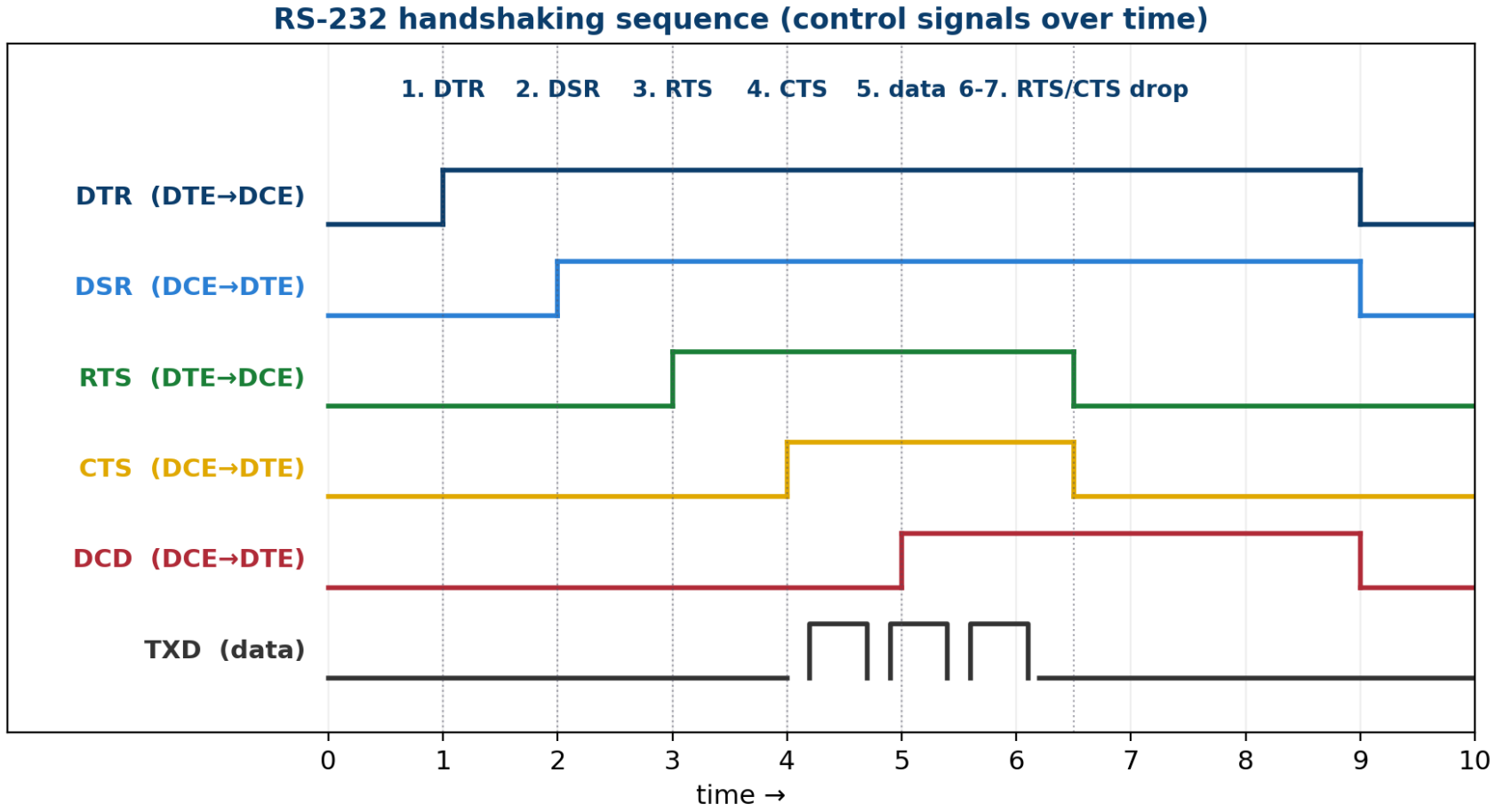


Three signal pairs: TXD/RXD (data), DTR/DSR (readiness), RTS/CTS (flow).

5. RS-232 Handshaking Sequence

1	**DTE asserts DTR** — Computer: “I am on and ready.”
2	**DCE asserts DSR** — Modem: “I am on and ready.”
3	**DTE asserts RTS** — Computer: “I want to send data.”
4	**DCE asserts CTS** — Modem: “Go ahead, send.” → data on TXD
5	**Data exchange** — TXD (send) / RXD (receive)
6	**DTE drops RTS** — Computer: “I have finished sending.”
7	**DCE drops CTS** — Modem: “Acknowledged.”
8	**Drop DTR/DSR** — Connection released.

5. RS-232 Handshake — Timing Diagram



Control signals are asserted in order before data flows on TXD.

5. Flow Control, Null Modem & Limitations

Flow control

Hardware (RTS/CTS): receiver lowers CTS to pause the sender.

Software (XON/XOFF): Ctrl-Q / Ctrl-S characters in the data stream.

Null modem

Connects two DTE devices directly.

Crosses TXD ↔ RXD and RTS ↔ CTS.

Often loops DTR ↔ DSR — no modem needed.

Limitations of RS-232

Short distance (≈15 m) and low speed.

Single-ended signalling → poor noise immunity.

Point-to-point only; bulky cables and large voltage swings.

Largely replaced by USB and differential standards (RS-485), still used in embedded/industrial devices.

6. Line Coding — Definition & Goals

Line coding is the process of converting a sequence of bits into a digital signal that can be transmitted over a baseband channel — mapping 1s and 0s to voltage or current levels.

Design goal	Meaning
No DC / baseline	Avoid drifting average voltage that confuses the receiver
Self-synchronisation	Enough transitions for the receiver to recover the clock
Error detection	Some codes can detect errors
Noise immunity	Resistance to interference on the line
Bandwidth efficiency	Use minimal bandwidth for a given data rate

7. Line Coding Schemes

Unipolar

Levels on one side of the axis.

Unipolar NRZ: 1 = +V, 0 = 0 V.

Simple, but DC component + poor sync.

Polar

Uses positive and negative voltages.

NRZ-L: level = bit value.

NRZ-I: 1 = transition, 0 = no change.

RZ, Manchester, Differential Manchester.

Bipolar

Three levels: +, 0, −.

AMI: 0 = 0 V; 1s alternate polarity.

Pseudoternary swaps the roles.

No DC, good bandwidth efficiency.

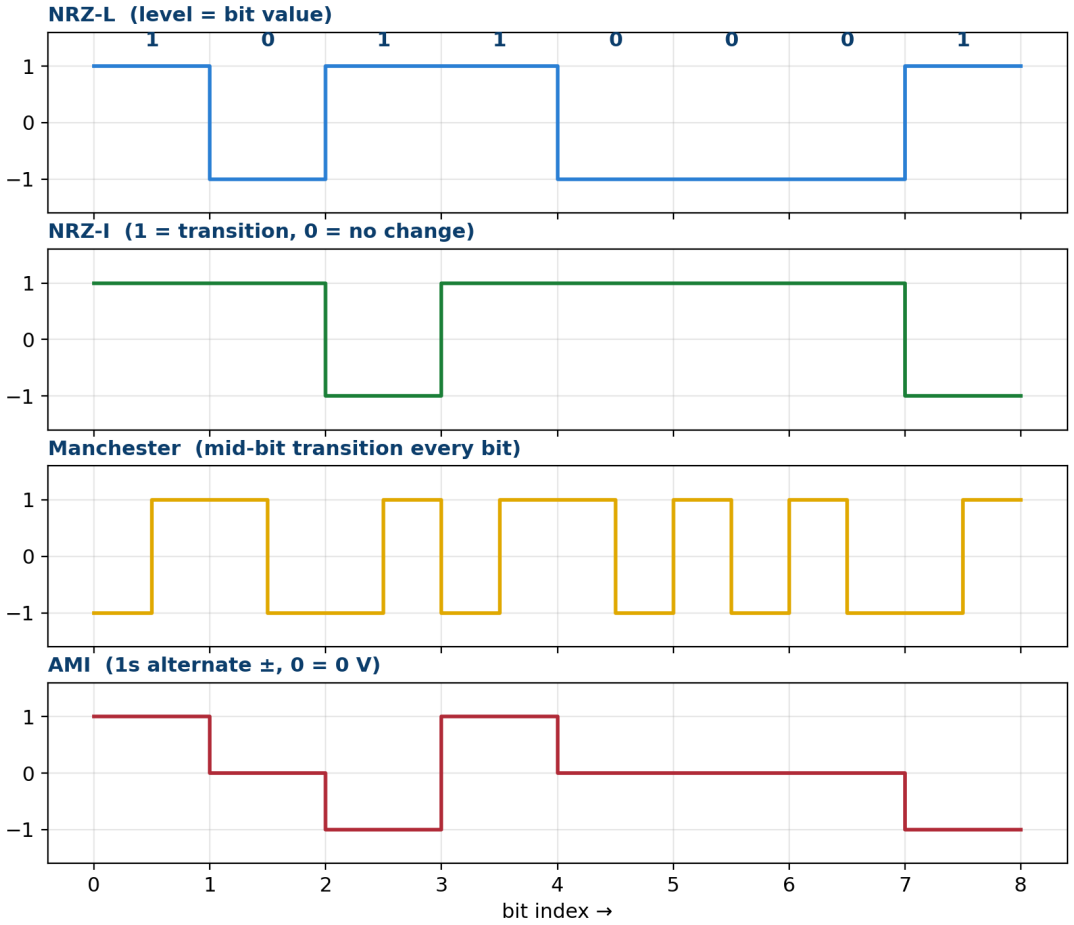
Block coding & scrambling

Block: m bits → n bits (4B/5B, 8B/10B).

Guarantees transitions, removes DC.

Scrambling: B8ZS, HDB3 for long distance.

7. Line Coding Waveforms



Same data bits (1 0 1 1 0 0 0 1) encoded with four different schemes.

8. Worked Example

Data bits: 1 0 1 1 0 0 0 1

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NRZ-L  :   +   -   +   +   -   -   -   +
NRZ-I  :   +   +   -   +   +   +   +   -     (1 = transition, 0 = no change)
Manchester : every bit has a mid-bit transition
AMI    :   +   0   -   +   0   0   0   -     (1s alternate, 0 = 0 V)
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Remember: NRZ-L follows the level • NRZ-I follows transitions • Manchester always has a mid-bit transition • AMI alternates polarity on 1s.

8. Comparison of Line Coding Schemes

Scheme	Levels	DC	Self-sync	Bandwidth	Typical use
Unipolar NRZ	2	Yes	No	Low	Simple / legacy
NRZ-L	2	Possible	No	Low	General digital
NRZ-I	2	Possible	Partly	Low	USB, Fast Ethernet
RZ	3	No	Yes	High (2×)	Legacy
Manchester	2	No	Yes	High	10Base-T Ethernet
Diff. Manchester	2	No	Yes	High	Token Ring
AMI	3	No	Partly	Moderate	T1 lines
4B/5B + NRZ-I	2	No	Yes	Moderate	100Base-FX

9. Summary

Modems

Convert digital ↔ analogue using ASK, FSK, PSK or QAM over lines such as telephone/cable.

RS-232

Serial DTE–DCE interface: voltage levels, DB-9/DB-25 pin-out, handshaking (DTR/DSR, RTS/CTS, DCD, RI).

Line coding

Turns bits into a digital signal. Good codes: no DC, self-synchronising, noise-resistant.

Families: unipolar, polar (NRZ/RZ/Manchester), bipolar (AMI), block coding (4B/5B), scrambling (B8ZS, HDB3).

Thank You

Questions & Discussion

Group 1 — CIT 3212 Computer Networks